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Diagnosis and treatment

Treatment of cognitive impairment in Parkinson disease☆

Tratamiento del deterioro cognitivo en la enfermedad de Parkinson

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Introduction

Mild cognitive impairment (MCI) and dementia are common disorders in Parkinson's disease (PD). The prevalence of PD with dementia (PDD) is over 80% at 20 years of follow-up. This leads the patient to lose autonomy, with high impact on their and their relative's quality of life.[1]

PD-MCI (cognitive performance below normal for the patient's age and educational level, which remains independent for activities of daily living [ADL]) increases the risk of conversion to dementia.[2,3] It is an heterogeneous entity, in both number and type of cognitive domains involved (attention, executive function, memory, visuospatial function and language). Subsequent cortical deficits (visuospatial function and semantic verbal fluency) appear to increase the risk of dementia.[2,4,5]

Neuropathology and biochemistry of cognitive impairment in Parkinson's disease

One of the causes of cognitive symptoms is the deficits in monoaminergic systems. The progressive loss of striatal dopaminergic innervation, in addition to causing the classic motor signs, associates executive dysfunction (such as difficulties in planning, attention, etc.). The noradrenergic deficit might also have a role in design and learning disorders due to the degeneration of the locus coeruleus. Finally, the deficit of acetylcholine, a neurotransmitter of major relevance in CI, is extensive in PD and PDD, even higher than in AD, and is involved in maintaining attention (involving indirectly executive function), memory loss, deficit of visuospatial function and visual hallucinations.[6]

The presence of neocortical and limbic LCs is the most important pathological substrate in the development of PDD. The typical pathology of AD (beta-amyloid and neurofibrillary tangles) occurs in half of the cases and might be involved in an early onset of dementia.[6]

This paper will describe the existing studies in the treatment of CI, PD and will give a practical vision of its management.

Methods

A bibliographic review and a review of the registry of the National Institute of Health of the USA has been carried out (www.clinicaltrials.gov) on pharmacological and non-pharmacological treatment (deep brain stimulation and non-invasive treatments: physical activity, cognitive stimulation, repetitive transcranial magnetic stimulation and transcranial direct-current stimulation).

Results

General approach

In a patient with PD and CI, we will first rule out non-disease causes that may worsen or cause degeneration, including drug adverse effects, depression, cerebrovascular disease, systemic or metabolic disorders. We will carefully review the medication, mainly recent modifications, since some treatments with anticholinergic effect (amantadine, trihexyphenidyl or oxibutitin for urinary symptoms), benzodiazepines and dopaminergic agonists might be involved. We must rule out reversible causes of cognitive impairment by analyzing thyroid function, folic acid, and vitamin B12. An abrupt onset would also lead one to think of other causes of degeneration. In some cases, a brain MRI might be useful[7] (Table 1).

Once secondary causes have been ruled out, we will consider non-pharmacological measures such as cognitive therapy and

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Table 1
Strategy for the treatment of mild cognitive impairment and dementia in Parkinson's disease.

1. Cognitive evaluation and study of potential concomitant depression
2. Careful review of the medication and adjustment: anticholinergic drugs, dopaminergic agonists, benzodiazepines
3. Blood test: vitamin B12, folic acid and thyroid function. Assess if the study of other metabolic/systemic impairment is necessary. Rule out urinary tract infection in cognitive impairment and/or associated psychosis
4. Brain MRI: non-diagnostic but it may show brain atrophy and rule out associated vascular pathology
5. In Parkinson's disease dementia: start treatment with anticholinesterase inhibitors. If possible, rivastigmine in patches
6. Physical exercise on a regular basis and intellectual activity

physical exercise (they can also be used preventively) and we will assess a potential benefit of the pharmacological treatment.

Pharmacological and surgical treatment

Pharmacological treatment includes strategies based on neurotransmitter deficiency.

Dopaminergic therapies can improve, mainly at the onset of the disease in some patients, aspects of executive function (mental flexibility or working memory), which would depend on frontostriatal associative circuits. On the contrary, other aspects of the executive function may worsen caused by an overstimulation of a less denervated limbic and orbitofrontal circuit.[8] In dementia, the potential negative cognitive effects are more evident, which can induce more confusion and hasten or worsen hallucinations. Reducing these agents, or replacing by any other type (eg, dopaminergic agonist to levodopa/carbidopa) may lead to clinical improvement. However, even though levodopa/carbidopa is best tolerated, side effects may continue.[7]

Parkinson's disease dementia

Anticholinesterase inhibitors. In a 24-week double-blind, placebo-controlled study of 541 patients with mild-moderate dementia, rivastigmine efficacy wass proved in the clinical global impression scale (CGIS) and neuropsychologically. The most common side effects were nausea-vomiting and worsening of tremor.[9] Subsequent studies found that improvement occurred in all aspects of attention[10] and in the reduction of visual hallucinations, frequent in PPD. In longer studies, the improvement lasted up to week 48, but less effective, being safe at week 76.[11]

The results with donepezil have been contradictory. In a randomized, double-blind, placebo-controlled study of 550 patients with mild-to-moderate PPD no cognitive differences were reported in CGIS between patients taking 10, 5 mg or placebo after 24 weeks. However, both doses of donepezil were proved to be effective in tests of global cognition, attention and executive function (secondary endpoint).[12]

The rest of the studies have included small samples, which may explain the poor or lacking benefit. In a 10-week double-blind, placebo-controlled study in 14 patients, donepezil improved cognition slightly with good tolerance.[13] In another double-blind, cross-sectional study with 2 periods of 10 weeks separated by a wash-out time no improvement was reported in cognition (primary endpoint) but it was in the CGIS (secondary endpoint).[14] Finally, Leroi et al., in 16 subjects randomized to placebo or donepezil, observed very slight improvement in memory, without other differences at a cognitive level. Side effects were more frequent in the donepezil group, with mild to moderate intensity.[15]

Finally, a in study with galantamine in PDD, efficacy was reported in cognition, hallucinations, anxiety, apathy and sleep compared to placebo, but it included a small sample (41 patients) and despite being controlled it was an open study.[16]

These drugs are well tolerated, with a dropout rate by 10–31% and side effects similar to those described in AD. The most frequent are nausea-vomiting, followed by hypersalivation, rhinorrhea and tearing (15%), and to a lesser extent postural hypotension, falls and syncope (10%). In some patients in the active group of the clinical trials there was worsening of parkinsonian symptoms, particularly in tremor, but no impact on motor UPDRS and total UPDRS.[17] Adverse effects appear to be less frequent in those using the rivastigmine patches.

Due to potential cardiovascular effects (hypotension, bradycardia) and, as a consequence, risk of syncope and falls, in clinical practice some specialists include an ECG and blood pressure measurement prior to treatment to rule out hypotension or underlying cardiac abnormalities that contraindicate their use. In general terms, anticholinesterase inhibitors pose an acceptable risk with no specialized supervision.

The Movement Disorders Society (2011 and 2013) reviewed the effectiveness of these treatments and showed that rivastigmine is effective and clinically useful. However there is no sufficient evidence for donepezil and galantamine but they are potentially useful based on their effectiveness and approval outside PD[18] (Table 2). In a recent meta-analysis it was reported that, although with mild effect, anticholinesterase inhibitors are effective in PDD, both in cognition and in behavioral impairment and ADL.[17]

An abrupt withdrawal may worsen cognitive and psychiatric symptoms. Therefore, it is recommended that this treatment be maintained long-term in patients with good response.

Memantine. It is an N-methyl-D-aspartate receptor antagonist whose mechanism would normalize glutamatergic dysfunction. In a randomized, controlled, small size (n = 25) double-blind 22-week clinical trial in PDD, there were no differences between the active and control groups. However, after withdrawal of the medication, patients who received memantine showed a higher degree of impairment. Thus memantine was proved to be beneficial to some extent.[19] In a larger (n = 72) (randomized, controlled, double-blind) study that also included patients with LB dementia, there was a significant benefit in the CGIS in favor of the active group but no improvement in secondary endpoints: The Mini-Mental State Examination (MMSE) and the neuropsychiatric inventory.[20] In a more recent study in a larger PDD and LB dementia population (199 patients), there were no differences at week 24 in the CGIS in the PDD active treatment group, but there were differences in the LB dementia group.[21] However, in another study with 51 PDD or LB dementia patients, memantine was effective in cognitive tasks in both groups, with a medium-long effect size.[22] Finally, it is worth mentioning an effect of this drug on survival in PDD and LB dementia patients, suggesting a potential modifying effect of the disease.

Memantine is well tolerated and it rarely causes adverse effects (as in AD). Therefore it poses no safety concerns. We can conclude that memantine at most has mild beneficial effects on PDD, mainly in global cognition and behavioral symptoms, but the data are contradictory. Therefore, it is deemed potentially useful but it does not show sufficient evidence to be considered clinically useful.[18] (Table 2).

Active clinical trials. Of the clinical trials described at www.clinicaltrials.gov, there is only one controlled, randomized, double-blind clinical trial in PDD, the SYNAPSE. This phase-II trial evaluates SYN120 safety, tolerability and effectiveness, a dual 5-HT6/5-HT2A dual antagonist in PDD patients treated with a stable dose of anticholinesterase inhibitors. These receptors involved in cognitive, mood and psychosis processes are located in

Table 2
Symptomatic treatment of dementia in Parkinson's disease.

	Dosage	Effectiveness and practical implications[a]	Considerations
Acetylcholinesterase inhibitors			
Rivastigmine (orally)	1.5 mg twice daily, progressively increased by 1.5 mg every 2 weeks to 6 mg twice daily	Effective. Clinically useful	–
Rivastigmine (patches)	4.6 mg/24 h for 4 weeks, then increase to 9.5 mg/24 h	Effective. Clinically useful	Useful patch formulation to prevent or treat gastrointestinal side effects. Available doses by 13.3 mg/24 h but no studies in PDD
Donepezil (orally)	5 mg/day for 4 weeks, then 10 mg	Insufficient but possibly useful evidence	FDA-approved 23 mg/day dose for moderate to severe AD, with a higher rate of adverse effects than standard dose. Under study for PDD
Galantamine (orally)	4 mg twice daily, increase to 8 mg twice daily for 4 weeks, increase to 12 mg twice daily at week 8	Insufficient but possibly useful evidence	–
Galantamine extended release (orally)	8 mg/day, increase to 16 mg at week 4 and to 24 mg at week 8	Insufficient but possibly useful evidence	–
NMDA receptor antagonists			
Memantine	5 mg/day for one week, increase by 5 mg weekly to 20 mg	Insufficient but possibly useful evidence	–

PDD: Parkinson's disease with dementia; NMDA: N-methyl-D-aspartate.
[a] Seppi et al., 2013 (http://display.mds.prod.titanclient.com/MDS-Files1/PDFs/EBM-Papers/EBM_NMS_Updated_15Jan2014.pdf).

the prefrontal cortex and the hippocampus. It is a 16-week trial in the recruitment phase.

There are 2 studies with donepezil in PDD. The first, MUSTARDD-PD, was a phase-III clinical trial in incipient PDD, but it was discontinued (NCT01014858). Finally, the NCT02415062 clinical trial compares high doses of donepezil (23 mg), FDA-approved in moderate-severe AD, versus 10 mg and is in the recruitment phase.

Deep brain stimulation. In 2009 it was performed for the first time in PDD with a target in the cholinergic nucleus basalis of Meynert (NBM). The purpose was to stimulate residual cholinergic projections. This patient was also implanted with bilateral electrodes in the subthalamic nucleus. After activating the NBM electrodes, the patient improved cognitively (attention, concentration and praxis).[23] There are 2 ongoing studies on PDD. The first, in the recruitment phase, would include the placement of electrodes in the bilateral internal globus pallidus, the deepest contacts being on the NBM (phases 1 and 2, NCT01701544). In the second (uninitiated), the electrodes are placed bilaterally in the subthalamic nucleus and in the NBM (phase 1, NCT02589925).

If this intervention were tolerated, patients might have a cognitive benefit, preventing the side effects of oral medications.

Parkinson's disease without dementia

Several clinical trials have studied the effectiveness of various drugs on cognition in non-demented PD patients (normal cognition and MCI). Only recently, studies have been conducted on PD-MCI population.

Anticholinesterase inhibitors. Linazasoro et al. reported improvement in 10 patients with executive dysfunction who received donepezil openly.[24] However, a double-blind controlled study in 69 non-demented patients treated with 16-week galantamine reported no significant difference from placebo and gastrointestinal side effects led to a high dropout rate.[25]

Atomoxetine (noradrenaline reuptake inhibitor). Two published studies have evaluated their response on PD cognition. The former showed improvement in the executive function in non-demented individuals, with good tolerance.[26] However, it was open and with a small sample ($n = 12$). In a randomized controlled clinical trial, Weintraub et al. assessed the depression response to atomoxetine in 55 patients (with MMSE > 14), with no improvement in this purpose. However they found improvement in global cognition (secondary endpoint) as measured by MMSE (screening test). Therefore, cognition was not fully evaluated.[27] Two further studies were conducted, but their results have not been published. In NCT01340885, a double-blind, phase-IV study in patients with non-demented PD 3 intervention groups were included (atomoxetine, rivastigmine and placebo) for 6 weeks. The NCT01738191-Atomoxetine Treatment for Cognitive Impairment in Parkinson's Disease was a double-blind, phase-II study in PD-MCI (diagnosed by Montreal Cognitive Assessment = 21–25, MoCA) randomized to atomoxetine or placebo for 12 weeks.

Monoamine oxidase B inhibitors. There are several studies in non-demented population with rasagiline, indicated to improve the motor symptomatology. A 12-week randomized placebo-controlled study in 55 patients showed a significant improvement in attention in the rasagiline group, with no further significant cognitive differences.[28]

Based on its potential beneficial effects, 3 subsequent studies have been conducted. Frakey and Friedman studied 50 patients with PD, no depression and MMSE > 23 for 6 months. It is a double-blind, placebo-controlled study (NCT01382342) reporting no improvement in cognition.[29] The other 2 evaluated the response to rasagiline in PD-MCI in a blind and placebo-controlled study, after 14 and 24 weeks respectively, but the results have not been published (NCT01497652, NCT01497652).

Finally, safinamide (recent approval for motor fluctuations in PD), which additionally has an inhibitory effect on excess glutamate release, has been the target of study in CI and non-demented patients (MoCA ≥ 26), but the results have not been published (NCT01211587).

With these data, we can conclude that no drug has been proved to be sufficiently efficient to be recommend in non-demented PD patients. However, some results have not been published. Some of the negative results might be explained by the small samples and by the heterogeneous PD-MCI. Thus, to reach firm conclusions we need studies with larger samples to differentiate the cognitive subtypes.

Non-pharmacological treatments

Due to the modest effect of the only approved treatment with anticholinesterase inhibitors in PDD and the lack of treatment for PD-MCI, we need other therapies that help improve and slow down CI.

Cognitive training

Cognitive training has been the most studied non-pharmacological therapy in PD. A recent meta-analysis showed small effect size on the executive function and a significant effect on processing speed. However, there was no improvement in global cognition, memory, visuospatial function, depression or quality of life.[30] Only 3 of the 7 included studies analyzed these changes in functionality and there was only improvement in one of them.

There is another randomized, but open-label controlled clinical trial whose results have not been published (NCT02225314), which evaluates the viability of cognitive training. It includes 3 active PD-MCI groups with 8-week computerized exercises: 2 cognitive training groups: Brain Fitness (auditory processing speed and accuracy) and Insight (visual processing and working memory) and an active control group (tests on literature, art and history). The primary endpoint is improving cognitive training tests and secondarily improving quality of life and memory.

Most of these studies have included non-demented populations, where it is not possible to know how many patients have CI and therefore it is unknown whether a significant impairment is being or not treated. On the other hand, its application might be deemed as preventive damage treatment. Therefore, one of the endpoints of this therapy should be to delay progression to PD-MCI and dementia. There is only one controlled study showing that both structured and randomized 6-week training would reduce the risk of progression from normal cognition to PD-MCI after one year compared to a control group that does not receive treatment. However, the control group does not undergo any therapy, so the placebo effect of the active group cannot be excluded.[31] There are no controlled studies of cognitive rehabilitation on an established CI intended to compensate for deficits through preserved cognitive skills and to optimize residual cognitive skills in cognitive impairment patients.

Physical exercise

InsSeveral studies, the effects of exercise on neuroplasticity in PD has been tested. Exercise would increase the expression of neurotrophic factors, brain flow, neurogenesis and immune response. Despite the potential benefits, few studies have been focused on cognition in PD.

Both, aerobic exercise and passive cycling, showed benefit in executive function for PD, but in uncontrolled studies.[32] In two studies, aerobic exercise and muscle endurance have been proved to be beneficial after 3 months[33] and 6 months[34] respectively. However, the control group remained with their routine activities, not controlling the placebo effect of the active group. More recently two types of physical exercise were compared: progressive resistance and one including stretching, balance, breathing and non-progressive strengthening. Both showed improvement in attention and interference. One of the strengths of this study is duration, since the patients received a 24-month therapy, but, on the other hand, there is no control group.[35] However, none of the previous studies evaluated changes in the patient's functionality and only one of them studied a potential effect on the quality of life, which was negative. There is an active clinical trial evaluating the benefit of 12-week exercise on cognition in PD-MCI, comparing 3 groups: 1. exercise of specific categories (e.g., balance, coordination), 2. aerobic exercise and 3. social contact (group interaction) as a control group (NCT02267785). Another ongoing clinical trial will evaluate the benefit of treadmill exercise, cognitive training, or both in combination on executive function in patients with balance problems (NCT01156714). Finally, the benefit of tango has been studied on PD cognition after 12 weeks of tango lessons, the results being contradictory. In one study improvement occurred in executive function in non-demented patients compared to the control group receiving lessons. In another controlled study with self-directed physical exercise in non-demented individuals there was no cognitive improvement, although it was measured only with the MoCA (screening test).[36]

Noninvasive neuromodulation

Noninvasive stimulation induces extended functional modifications in the cerebral cortex, modulates brain plasticity, and is effective in treating the motor symptoms of PD.

Repetitive transcranial magnetic stimulation. Only 2 controlled studies have evaluated efficacy in cognitive symptoms with high frequencies on the left dorsolateral prefrontal cortex (DLPFC) in non-demented PD patients for 10 days. However, the main endpoint of study was depression and cognition a secondary endpoint. The first (Boggio et al.) showed a small effect size in attention and executive function after 8 weeks, as well as improvement in depression. Therefore, it cannot be ruled out that the cognitive benefit was secondary to the improvement in mood. The second, found improvement in depression after 30 days, but not cognitive improvement or in ADLs.[37] In addition, an ongoing study is focused on PD-MCI patients receiving high frequency transcranial magnetic stimulation for 2 weeks compared to a placebo group (sham) (NCT02346708).

Transcranial direct current stimulation. Doruk et al. studied cognitive improvement in non-demented PD patients by stimulating right or left DLPFC for 10 days compared to a placebo group. In the comparison with placebo (sham), when stimulating both right and left DLPFC, only one executive function test (trail making test part B) had improved after one month of follow-up, and there was no benefit over visuospatial function.[38] The other 2 studies stimulated DLPFC, also for 2 weeks. In the first, the left DLPFC was stimulated and after 16 weeks improvement was reported in several attention and executive function tests, but not in memory, visuospatial function, or quality of life.[39] Finally, Manenti et al. studied the combined effect of this therapy on DLPFC from the contralateral side to the most affected motor side, with physical exercise for 10 days in PD-MCI compared to placebo stimulation (sham) and the same physical exercise. Three months later, patients who received the actual treatment with transcranial direct current stimulation improved frontal-subcortical function tasks, but not visuospatial function or denomination.[40] Despite being a combination therapy study, the lack of a control group for physical exercise does not allow us to know its specific benefits. As in the other therapies, noninvasive stimulation requires further study to assess the long-term effect and with larger samples.

Conclusions

CI is very widespread in PD and can lead to losing quality of life, patient's autonomy and increased caregiver burden. To date, we have only available symptomatic therapies (anticholinesterase inhibitors) for the treatment of dementia, with modest benefit, but no drugs have been approved for PD-MCI. There are many drugs under study (Table 3), which might have a beneficial effect on cognition, but most have been studied in small samples, have not been evaluated in dementia and only a few in PD-MCI. It is likely that,

Table 3
Other therapies under study for the treatment of cognitive impairment in Parkinson's disease.

Pharmacological therapies
SYN120, a 5-HT6/5-HT2A dual serotonin antagonist (PDD)*
Atomoxetine (noradrenaline reuptake inhibitor)
Monoamine Oxidase B Inhibitors: Rasagiline and Safinamide
Non-pharmacological therapies
*Deep Brain Stimulation (NBM) (PDD)**
Cognitive training
Physical exercise: aerobic, resistance and tango
Noninvasive neuromodulation
Repetitive transcranial magnetic stimulation
Transcranial Direct Current Stimulation

PDD: Parkinson's disease with dementia; NBM: cholinergic nucleus basalis of Meynert.
* All studies are on demented population except for two that are in demented population.

as in PDD, those PD-MCI patients closer to PDD (with more posterior cortical abnormalities), could benefit more. However, this differentiation between PD-MCI groups has not been carried out. On the other hand, drugs such as atomoxetine might be beneficial and although the results of a completed study have not been released yet, further trials in PD-MCI are required. Deep brain stimulation of NBM might be promising, restoring cholinergic dysfunction and thus preventing from oral agents with increased risk of systemic effects, although it would involve an invasive intervention. Non-pharmacological therapies (Table 3) may be useful and complementary to pharmacological therapies, although the data are inconclusive. Controlled studies should be conducted to eliminate confusing factors such as the placebo effect or socialization, focused on patient functionality, quality of life, and long-term benefit. Studies focusing on rehabilitating existing cognitive impairment are also required.

In addition to symptomatic treatment, we need disease-modifying therapies that prevent, stop or delay CI. In this sense, just as immunotherapy is used to reduce the accumulation of pathological proteins in AD, the reduction of alpha synuclein present in PD and especially in PDD by immunotherapy might be an effective treatment.

Conflict of interests

The author reports no conflict of interest regarding the contents of this article.

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